Jimmy Zheng

University of Waterloo • Data Science

Portfolio Website | LinkedIn/Jimmy | GitHub/Jimmy

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EDUCATION

University of Waterloo, Waterloo, Ontario — *Sep 2024 - Aug 2028*

- Bachelor of Mathematics, Data Science — GPA: 3.7/4.0
- Relevant Coursework: Object-Oriented Programming, Compilers, Algorithm Design & Data Structures, Linear Algebra

TECHNICAL SKILLS

- Languages: Python, JavaScript/TypeScript, Java, Go, C, C++, HTML, CSS, SQL, Dart
- Frameworks/Libraries: React, React Native, Vue.js, Node.js, Express.js, Next.js, NestJS, Flutter, .NET
- Tools: Postman, Docker, Jenkins, Stripe, Git, Bitbucket, n8n, Cloudflare, Jira, Cursor, Supabase, Vercel

WORK EXPERIENCE

Software Engineer, Co-op — *Sep 2026 - Present*

Kinaxis Inc, Ottawa, Ontario (C++, Visual Studio, Natvis, GoogleTest (gtest), Jenkins, CI/CD, Bitbucket, Jira)

- Developed custom **Natvis** debugger visualizations for complex **C++** domain objects, exposing key identifiers and improving **debugging efficiency** while preserving release-build behavior.
- Built reusable **C++ b-map container utilities** and implemented automated **GoogleTest (gtest)** and **KUnit tests** to validate functionality and improve regression coverage across Core-Algorithms components.
- Enhanced **Jenkins CI/CD reporting** by integrating **Performance Test** and **SQM Test**, improving build-health visibility and simplifying failure diagnosis.

Full-Stack Developer, Co-op — *Jan 2026 - Apr 2026*

Hanov Solutions Inc, Waterloo, Ontario (Python, Go, Vue.js, Node.js, TypeScript, SSR, pSEO, SQLite, Stripe)

- Engineered and deployed core infrastructure for eh-trade, a momentum-based stock analytics SaaS platform, transforming **Python ML** trading models into a production-ready full-stack system serving real-time stock screening.
- Optimized **Vue 3** and pair analysis interfaces by refactoring **data-heavy frontend** modules, reducing redundant **API** and **RPC** calls, and implementing **lazy loading**, cutting load times by **30%** across desktop and mobile.
- Built a scalable **programmatic SEO** (pSEO) system using **Go**, implementing **server-side rendering** (SSR) for dynamic stock pages, automated **meta/OG tag** generation, **cache-backed data** prewarming, and **sitemap** integration, enabling efficient **indexing** and fast crawl performance across **500+** earnings pages and hundreds of comparison URLs.
- Developed secure **authentication** and **subscription** infrastructure using **TypeScript**, **Go**, and **Stripe**, enabling recurring billing, gated premium features, and full user lifecycle management from onboarding to subscription control.
- Used **AI-assisted** development workflows (Cursor, Claude) to speed implementation of full-stack features including **charting** improvements, **watchlist** performance, **subscription** flows, and analytics tooling across **Vue.js, TypeScript, Go,** and **Python**, increasing feature throughput by **40%** without compromising review quality or system reliability.

Software Innovation Developer, Co-op — *May 2025 - Aug 2025*

Emerson, NI, Shanghai, China (C++, Python, LabVIEW SDK, AST, n8n, REST API, RAG, Docker, React)

- Assisted in developing a LabVIEW-to-native-code compiler using **C++ and Python**, enabling **graphical programs** to be translated into machine code. Improved compilation speed by **30%** and reduced runtime errors by **20%**.
- Built automated workflows with **n8n** and **Docker** to generate and publish circuit documentation, reducing manual preparation time by **70%** and ensuring consistent formatting across all projects.

Software Developer, Part-time — *Feb 2024 - Aug 2025*

VolunTrack.Org, Toronto, Ontario (Python, React, JavaScript, Jest, Node.js, Express.js, Postman, HTTP)

- Designed and executed structured **API** test cases in **Postman** for **happy-path** flows, **required-field** checks, and **type validation**, verifying correct status codes, response structures, and backend behavior across core endpoints.
- Implemented automated testing using **Python unit tests**, **Jest**, and **visual regression** testing, identifying and resolving **15+** bugs before production release and improving overall system reliability **by 50%.**
- Extended test coverage to **authentication** failures, **nonexistent-resource** scenarios, **duplicate submissions**, **rate-limit** behavior, improving backend response accuracy to **95%+** and reducing integration defects across the web application.

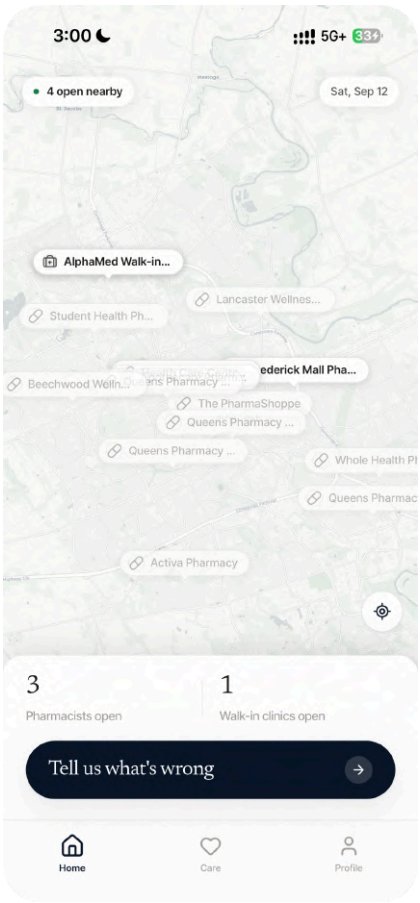
Portfolio

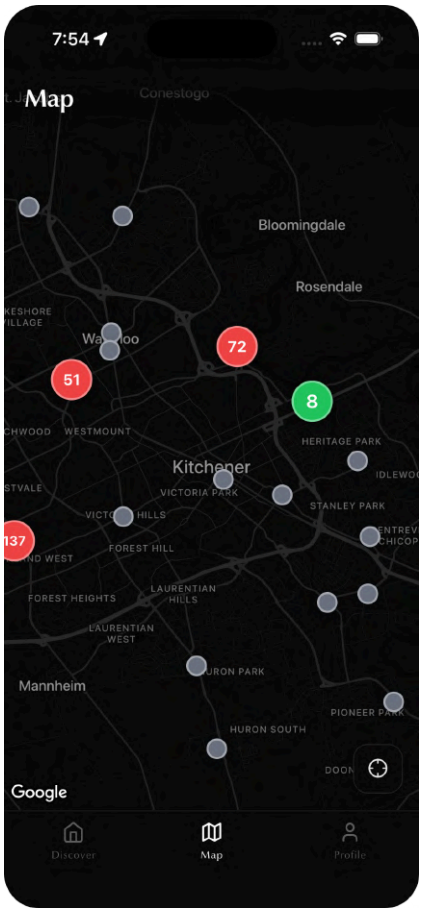
Jimmy's Portfolio
Jimmy's GitHub

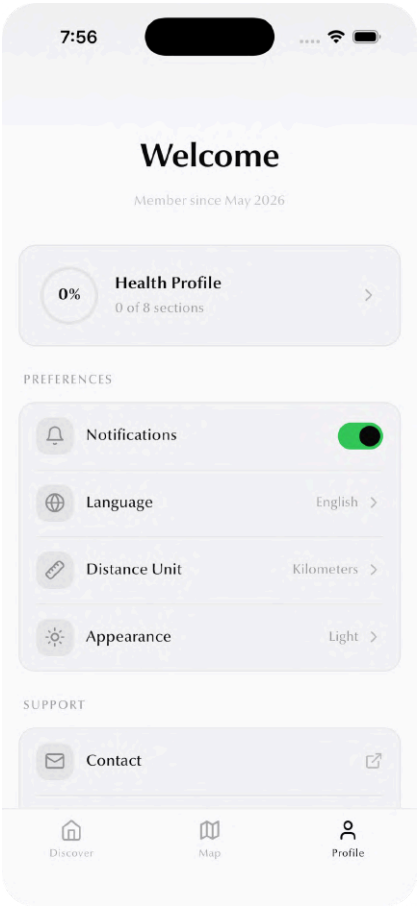
Fountain Health App (App Store/Google Play) *Feb 2026 - Present*

➢ **Tech Stack:** Flutter, Dart, .NET, NestJS, Supabase, PostgreSQL, Cloudflare, Postman, Xcode, Android Studio, iOS Simulator, Android Emulator, REST APIs, Apple App Store ASO

- Built a cross-platform healthcare app for Waterloo and Kitchener clinics, enabling patients to view **real-time wait times** and **join online queues,** reducing unnecessary waiting time by **50%**.
- Developed **10+** responsive **Flutter screens** and reusable UI components with light/dark theme support, improving consistency across **iOS** and **Android** devices.
- Built an **offline-accessible** shared patient profile feature, allowing users to save medical information once, reuse it without network access, and reduce repeated on-site form completion by approximately **60%**.
- Tested **15+ REST API** flows and **8+ iOS/Android device** configurations using **Postman**, **Xcode**, and **Android Studio**, improving **frontend-backend** reliability and cross-device compatibility.
- **Optimized Apple App Store ASO** by planning app keywords, screenshots, subtitles, and store positioning to improve organic discovery and free user acquisition.
- Improved app reliability and **loading performance** by working with **NestJS** backend services.







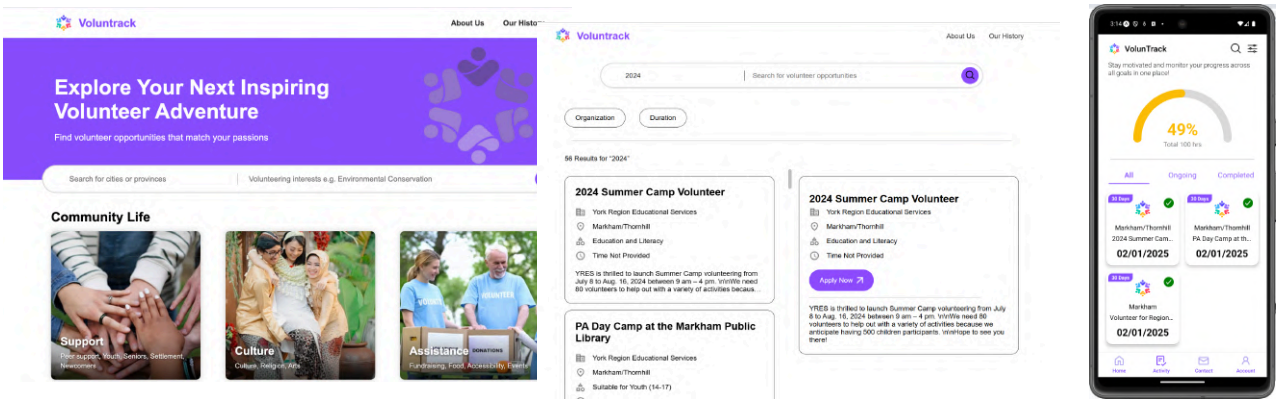
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VolunTrack Web & Mobile App (Repository) *Feb 2024 - August 2025*

➢ **Tech Stack:** React, React Native, JavaScript/TypeScript, Python, Node.js, Express.js, PostgreSQL, Jest, Postman, REST APIs

- Designed and developed **web** and **mobile pages** using **React** and **React Native** for a non-profit volunteer platform, helping high school students discover relevant volunteer opportunities.
- Built and maintained a **PostgreSQL-backed** volunteer opportunity database, regularly updating new listings and implementing a **search bar** that reduced search time to under 2 seconds.
- Implemented **API** and automated testing with **Postman**, **Python unit tests**, and **Jest**, validating core user flows, catching **15+** bugs pre-release, and improving application reliability by **~50%**.



Personal Website (Repository) *Feb 2026 - Present*

➢ **Tech Stack:** React, Vite, Next.js, Vercel, Express.js, OpenAI API, Vector Embeddings, RAG, Interact.js, Google Maps API, WebP, CORS

- Deployed and maintained a responsive portfolio website on **Vercel** using **React**, **Vite**, and **Next.js**, with reusable components, interactive UI sections, and route-based page navigation.
- Built a custom **grid-based drag-and-drop** system with **Interact.js**, enabling users to rearrange homepage components and photography page images through interactive swapping.
- Implemented an **Express.js RAG API** using **OpenAI API** and **vector embeddings**, including **CORS** handling, **`/api/ask`**, **`/api/health`**, and **`/api/reindex`** endpoints.
- Improved mobile and desktop performance for **300+** WebP images with **lazy loading**, **thumbnails**, **async loading**, and **fetchPriority**, reducing loading time by **80%**.

